

HSBC Online Test – Mini Practice Set

■ Aptitude (Quant + Logical + Verbal)

- 1. Number Series: Find the next number → 2, 6, 12, 20, 30, ?
- 2. Probability: A bag has 4 red, 5 blue, 3 green balls. Probability of drawing 2 blue balls?
- 3. Time & Work: A does a job in 10 days, B in 15 days. How many days together?
- 4. Permutation: Number of 4-letter words formed using letters of "HSBC" without repetition.
- 5. Ratio/Proportion: If 3 mangoes = 2 apples, 6 apples = 4 bananas, find mango:banana ratio.
- 6. Logical Puzzle: 5 friends (A, B, C, D, E) sit in a line. A is left of B, C is not at any end, D is left of E. Who is in the middle?
- 7. Blood Relation: A says, "The man in the photo is my father's son." Who is in the photo?
- 8. Data Interpretation: A shop's sales: Jan – 200, Feb – 250, Mar – 300. Find % increase Jan → Mar.
- 9. Sentence Correction: "He don't like coffee." (Fix grammar).
- 10. Synonym: "Prudent" → (a) Careless (b) Wise (c) Harsh (d) Proud

■ Coding (Solve at least 2 fully)

- Q1. String Problem: Check if a string is a palindrome (ignore spaces, case).
- Q2. Array Problem: Find the second largest element in an array.
- Q3. Recursion Problem: Find the nth Fibonacci number using recursion.
- Q4. Hashing Problem: Given an array, check if any element appears more than once.
- Q5. Sorting/Searching Problem: Implement Binary Search on a sorted array.

■■ CS Fundamentals (Quick Revise Q&A;)

- DBMS: 1. Difference between INNER JOIN and OUTER JOIN?
- DBMS: 2. SQL Query: Find all employees with salary > 50,000 from table Employees.
- DBMS: 3. What are ACID properties in transactions?
- OOP: 4. Explain Polymorphism with a real-life example.
- OOP: 5. Difference between abstract class and interface?
- OS: 6. What is a deadlock? Name its 4 conditions.
- OS: 7. Difference between process and thread?
- CN: 8. Which layer of OSI model handles routing?
- CN: 9. Difference between TCP and UDP.
- CN: 10. What is DNS used for?